CRDTs and Databases: A Hands-On Tutorial

Part 4 — Open challenges

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Open challenges

CRDVs show that conflict-free replication is achievable on mainstream SQL systems using widely available features but exposing the limits of the infrastructure.

Integration

Balancing encapsulation against the flexibility that makes merge semantics expressible.

Deployment

Optimizers and replication mechanisms that suit stacked views, logical clocks, and many sites.

Transactions

Guarantees that span sites, and the interfaces needed to compose them.

Correctness

Testing concurrency once part of it lives in application code.

Integration

- CRDVs are encapsulated with rules and views, so the application sees ordinary tables and writes ordinary SQL.
- The abstraction nevertheless **leaks**. The developer is not shielded from mistakes in the metadata, in the resolution rules, or in the interaction between them.
- That same exposure is precisely what allows **application-specific merge strategies** to be expressed.

Open problem

Where to place the boundary between encapsulation and flexibility. A fully sealed abstraction returns us to a fixed catalogue of types. A fully open one leaves correctness to the application.

Directions

- declaring conflict-free tables and their resolution rules as **first-class schema objects**, rather than as a convention over views
- checking that a rule is well formed, for instance that it is deterministic and independent of the order in which concurrent versions are presented

Deployment

Query optimization

- stacked views over logical clocks are not a workload current optimizers were tuned for
- a single index on the timestamp accelerates only equality. Range operators compare component by component from the left, so $[0, 1]$ appears to precede $[1, 0]$ when in fact the two are concurrent
- geometric encodings answer containment with one index, at the cost of a considerably larger index
- planning time grows with nesting

This motivates optimizer and access-path support for **causality as a first-class predicate**.

Replication and topology

- existing replication infrastructure was not designed for the scale that conflict-free replication makes both achievable and desirable
- with a fully connected topology the overhead is proportional to the number of sites, and write throughput stops scaling beyond four
- a ring restores throughput, but increases replication delay and makes availability depend on reconfiguring the ring

This motivates **larger and more complex overlays**, with arbitrary forwarding rules.

Transactions

CRDVs inherit the local ACID guarantees of the underlying system. Guarantees that **span sites** are out of scope, and are the most demanding direction.

- the hard question is what should happen when **strong and eventual transactions touch the same data**
- a resolution rule such as *strong-wins* may be the answer, which would keep the combination expressible in the same framework

Related work

D. Pereira, F. Pereira, R. Lopes, N. Faria, P. Almeida, and J. Pereira. *Towards Abstractions for Composable Transactional Isolation.* Fourth International Workshop on Composable Data Management Systems (CDMS) @ VLDB, 2026. **Friday morning!**

Correctness

- Building on CRDVs moves part of the responsibility for concurrency into libraries and application code, namely **which versions are visible** given their vector timestamps.
- Concurrency is notoriously difficult to get right and hard to test.
- Current transactional isolation checkers assume read and write operations over opaque values. They **cannot handle arbitrary data types** whose correct outcome depends on merge semantics.

What is needed

Configurable and extensible checkers, in which the expected outcome of concurrent operations is a parameter rather than a fixed rule.

Related work

M. Barros, A. Cunha, J. Pereira, and E. Kang, *Reasoning about Transactional Isolation Levels with Isolde*, arXiv:2604.00159, 2026.

Wrap-up

- CRDTs give a principled account of what can be replicated without coordination, in two main approaches and their variants.
- Expressing them as **views over a replicated history** brings merge semantics inside the query processor, which gives both application-defined rules at runtime and ordinary query optimization.
- Can be deployed today.

Artifacts

- github.com/nuno-faria/crdv
- github.com/dbr-haslab/crdts-and-databases

Thank you!

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